

ANTUMBRA

A Reputation-Anchored Settlement Network for the Human-Machine Economy

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Abstract

ANTUMBRA is a settlement network built for a problem that no existing chain treats as a protocol primitive: verifiable trust between humans and machines. Millions of software agents already buy, sell and pay on behalf of people, over rails built on fully transparent stablecoins; those rails say nothing about whether an agent is reliable, who answers for it, or where its budget ends. Dominant proofs of humanity engrave body data into global registries, and the coins that are actually private remain slow or unreadable to regulators. ANTUMBRA aims at that square: no deployed chain combines the five properties it requires, and each of the nearest neighbors renounces at least two. It couples a CPU-mined BlockDAG with two-second blocks and private-by-default transactions, a finality committee of fifty-five seats drawn by reputation rather than capital, non-biometric human identities, agents that are accountable by construction, and a native three-level disclosure mechanism. The scarce resource the network mints is consensus reputation, Kleos: three layers of which the heaviest is time itself, which cannot be bought. Above the money, a machine economy pays agents by verified result and efficiency through bounded Mandates, a labor exchange with a contest window, and a second reputation clock for machines, with diversity caps the deterministic simulation attacks with ten thousand clones. An immunity layer, the Thymus, pays for the chain's own breakage on replayable proof and freezes upgrades, never payments. The evolution layer, the Corona v2, is a state machine over four rails of change, merged by a diverse predicate under human silence or human vote, never by unanimity. The money supply is capped at 16,180,339 ATU, the golden ratio scaled by ten million, issued over thirty-four four-year eclipses and closed exactly at year 136. The specification was attacked by its own deterministic simulation before being written down as final: one flaw was found and four corrective rules closed it, the engine is public code, and the rules hold on a sweep of twelve seeds. A first independent review of version 1.1 returned twelve findings; version 1.2 closed them. Version 1.3 closes the machine-economy gaps of the second review: uniform authority encoding, the Mandate grammar, the immunity market, the rails, the predicate, and the honest metrics, MTTD, MTTC and MTTR, that replace the word perfect. This document is the public reference specification for the network.

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1 Introduction

A blockchain earns its existence only by serving a structural, growing and badly served demand. Four currents converge in 2026 on the same point, and no deployed chain occupies it: the nearest neighbors, reviewed in Table 2, each renounce at least two of its requirements. The first is agent commerce. Coinbase revived HTTP status code 402 with the x402 protocol, Stripe and Coinbase contest the machine-to-machine payment rails, and four rival standards share the same territory. All of these rails share three blind spots: they are denominated in stablecoins, they are fully transparent, and they carry no portable trust. An agent pays on them; nothing says whether it is reliable, for whom it acts, or who answers for it. Yet a merchant cashing out an unknown agent, an accountant auditing a fleet of agents, and a client delegating a budget all need the same thing: to know who they are dealing with.

The second current is proof of humanity. Bots saturate the web to the point that platform defense has become the separation of persons from programs. The dominant answers are biometric: iris scans for one project, biometry woven into consensus for another. Each accumulates the same debt, a body measurement that is irreversible, centralized at collection even when the proof is decentralized. Once an iris template is engraved somewhere, there is no second chance; the perfect registry of future leaks is already written. The third current is regulatory. MiCA and the FATF Travel Rule keep sustained pressure on private coins, while Zcash demonstrates the practicable path: a regulator who looks at its mechanism understands it, shielded pools, viewing keys, explicit disclosure. Readability, not transparency, is the acceptance criterion. The fourth current is instantaneity. A counter payment must become irreversible in seconds; the existing private chains require tens of seconds to minutes of real finality.

Table 1: The four demands of 2026 and the empty square at the center.

Demand in 2026	Dominant offer	Blind spot left open
Agent payments (x402, MPP, ACP, AP2)	HTTP rails on transparent stablecoins	portable trust, privacy, a designated answerable party: none
Proof of humanity	biometrics (iris, face, veins)	non-biometric personhood, with no body registry
Sustainable privacy (MiCA, Travel Rule)	Zcash: selective disclosure; Monero: opacity	native privacy with provable compliance
Quasi-instant private payment	Kaspa (fast, transparent); Monero (private, slow)	fast, private, finalized in seconds, on egalitarian hardware

The center of Table 1 is empty, and not for technical reasons. It is empty because each ecosystem optimized a single dimension: agent rails optimized throughput, biometrics optimized anti-sybil, Zcash optimized regulatory legitimacy, Kaspero optimized speed. None of them treats trust, meaning who acts, since when, answerable up to which human, as a primitive of the protocol. The network that installs trust there must combine five properties usually considered incompatible: speed, privacy by default, provable disclosure, processor egalitarianism, and a reputation that capital cannot fabricate. The thesis of this paper is that these properties reinforce one another as soon as one of them, reputation anchored in time, is treated as the foundation.

Table 2: The neighbors of the square, and what each renounces. The reference here is honest: rings and fast DAG ordering are inherited arts, not inventions of this paper.

Chain	Closest strength	What it renounces of the five properties
Monero, NERVA	the reference of private payment	speed (minutes of finality), agent identity, disclosure levels
Zcash (Orchard)	readable selective disclosure	egalitarian CPU mining (Equihash ASIC territory), agent identity, speed
Kaspero	fast DAG ordering at scale	privacy entirely, disclosure, reputation
Xelis	encrypted amounts on a DAG	ring anonymity (no decoys), reputation, CPU-only mining
Penumbra, Namada	shielded multi-asset, ZK disclosure	CPU mining, egalitarian hardware, reputation as consensus
Aztec, Aleo	private general computation	every proof circuit is maximal technique and maximal audit surface; CPU mining
ANTUMBRA	the five properties together	general masked contracts, honestly deferred (Section 13)

The monetary and consensus numbers that recur throughout this document are gathered here for reference. Each is justified in the section indicated.

2 Design Overview

ANTUMBRA separates cleanly what most chains pile up: settlement, finality, identity and disclosure. The settlement layer, at the bottom, is a CPU-mined BlockDAG that orders parallel blocks every two seconds and carries private-by-default transactions: this is the Veil. The finality layer, the Ring, is a committee of fifty-five high-reputation identities that sign checkpoints every four seconds and make any payment irreversible in under six seconds. The identity layer distinguishes humans, the Embers, from agents, the Ciphers, with their admission and accountability rules. The reputation layer, Kleos, feeds the Ring, locks sponsorships and weights peer testimony. Around the whole, the Lumen, selective disclosure, forms the ring of light that gives the network its name: a shadow faithfully kept, proofs on demand.

The nomenclature is stable across the document; each name designates one precise part and only one. The reader may return to Table 4 at any time. The internal names are working codenames, chosen to keep the document readable; they may be adapted at implementation time. Only ANTUMBRA and its ticker ATU are proposed as a definitive commitment, verified free of collision on the blockchain and crypto namespace on September 6, 2026. One layer of Table 4 sits above the others and outside the stack

Table 3: Reference magnitudes of the network.

Quantity	Value	Meaning
Monetary cap	16,180,339 ATU	the golden ratio scaled by ten million; the same proportion already drives RandomX
Emission	34 eclipses of 4 years	each eclipse emits 61.8 % of the previous one; the cap is reached exactly at year 136
Blocks	2 seconds	near-immediate inclusion; the DAG absorbs parallel blocks
Economic finality	under 6 seconds	checkpoints signed by the Ring, quorum of 37 out of 55 seats
Privacy	private by default	rings of sixteen, committed amounts, one-time addresses, Tor by default
Disclosure	selective, three levels	per transaction, per bounded auditor, per compliance proof
Reputation	Kleos, 0 to 100	Deed 40, Echo 30, Tenure 30; non-transferable, corroded by time
Premine	0 %	everything is mined or governed; nobody starts with an advantage

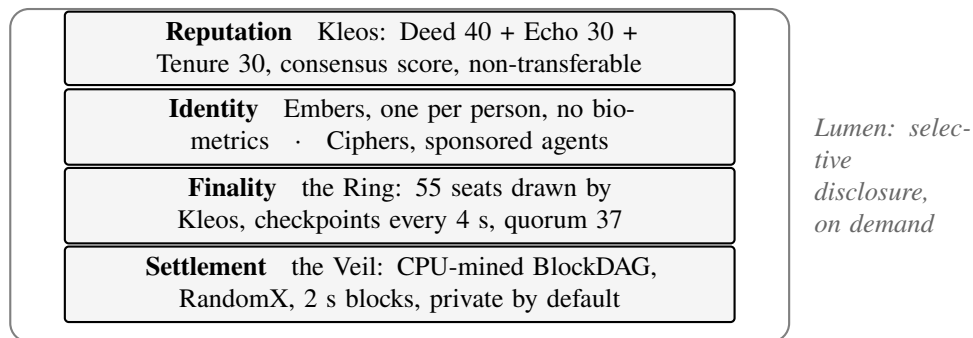


Figure 1: The four layers of ANTUMBRA inside the Lumen, the selective disclosure ring. Each layer depends only on the layers below it.

discipline: the Corona, Section 15, machine governance that reads every layer and modifies none of them without the human chambers.

Table 4: Reference nomenclature.

Name	Nature	Role in one sentence
ANTUMBRA	the network	private settlement BlockDAG, CPU-mined
ATU	the currency	16,180,339 units, divisible by 10^8
the Veil	privacy	rings of sixteen, committed amounts, one-time addresses, Tor by default
the Lumen	disclosure	hierarchical viewing keys and compliance proofs, on demand
the Ember	human identity	one per person, non-biometric, one vote in governance
the Cipher	agent identity	sponsored by an Ember, with a revocable spending perimeter
Kleos	reputation	three-layer score, consensus, non-transferable, non-purchasable
the Ring	finality committee	55 high-reputation seats, checkpoints every 4 seconds
RAF	finality mechanism	Reputation-Anchored Finality
the Corona	machine governance	AI senators that propose, red-team and implement changes under human veto (Section 15)
the Vault	deployment custody	threshold signature of 9 hardware fragments, 7 needed, no whole key anywhere
era	protocol time	six months; the unit of reputation, rotation and decay
eclipse	emission time	four years, eight eras; 34 eclipses lead to the cap at year 136
Mandate	money contract	an output whose release obeys a predicate declared at creation
the Machine	logic contract	deterministic virtual machine on public state only, after genesis

Two principles govern the assembly. First, each layer depends only on the layers below it: settlement ignores reputation, finality reads Kleos but never writes it, identity builds on both without modifying either. That stacking discipline is what keeps every part auditable in isolation and replaceable without rewriting the rest. Second, the value loop: reputation protects finality, finality makes payments instant, instant payments serve agents and merchants, and their honorable behavior enriches reputation. The network is designed so that its security and its utility feed each other instead of taxing each other.

3 The Settlement Layer: a CPU-Mined BlockDAG

The first wager is egalitarian. RandomX, the algorithm designed so that one ordinary computer is one vote, produces blocks that no specialized machine produces better at equal cost: no dedicated circuit, no farm of graphics cards. Miners run it in fast mode, 2,080 MiB of memory; verifying nodes may run light mode, 256 MiB of cache and no dataset, a size its author chose deliberately so that specialized hardware gains nothing by verifying: the light-mode memory-time product was sized to close exactly that door. Emission and block production stay within reach of anyone who owns a laptop, the ethos of NERVA carried to its conclusion. The constant $0x9E3779B9$, the integer truncation of the golden ratio in binary arithmetic, already drives the generation of RandomX superscalar programs; the same proportion that will distribute the currency already works inside the engine that will mine it, a mnemonic accident the paper claims as identity, not as argument.

The second wager is structural. A linear chain must choose between short blocks, which multiply orphans, and long blocks, which slow confirmation; a DAG orders parallel blocks instead of rejecting them. At a two-second cadence, a linear proof-of-work network would lose an unsustainable share of its blocks; a fast-converging DAG integrates them into its partial order and accumulates the work of all of them. Speed stops being a trade-off against egalitarianism: it becomes its product. The ordering layer belongs to the GHOSTDAG family, whose public literature and existing implementations provide the operational definitions of convergence and anticone.

Pruning deserves its own paragraph, because it conditions the entire social contract, and in a ring chain it is not free: spent outputs remain decoys for everyone else's privacy, so the output keys, their

Table 5: Parameters of the settlement layer.

Parameter	Retained value	Justification
Mining algorithm	RandomX: fast mode 2,080 MiB, light mode 256 MiB	proven by years of attack; the light-mode cache is sized so verification resists specialization
Block cadence	2 seconds	near-immediate inclusion; the DAG absorbs the orphan rate
Block ordering	GHOSTDAG-family coloring, cluster parameter $k = 8$	security grows with block volume, not only depth; k is calibrated for the two-second cadence
Flow anonymity	ring of 16, masked amounts	equals the strongest ring deployed in the lineage (Monero since 2022); a floor to raise, not a ceiling
Signature scheme	MLSAG over Ed25519, Keccak-256	the lineage construction, implemented and cross-validated; CLSAG is the post-audit upgrade candidate
Transport network	Dandelion++, Tor for transaction injection	IP-address analysis is the number one practical flaw of private chains
Finality	Ring checkpoints (Section 5)	economic irreversibility in under 6 seconds
Target throughput	100 tx/s sustained, 300 tx/s burst, on the reference node	an engineering budget with a measured exit criterion (Table 6)
Storage	pruned DAG, decoy pool retained forever	the nullifier set and output keys are consensus data; the growth budget is stated in Table 6

commitments and the nullifier set are consensus data forever, roughly ninety-six bytes per output. What prunes is the rest: the ring signatures, the range proofs and the transaction bodies beyond the recent window, which no later verification re-reads. A network whose nodes required a disk bay would exclude exactly the participants it claims to serve; the growth budget of Table 6 keeps that promise measurable. Dandelion++ stem propagation and Tor transport close the door on IP-address analysis, which remains the great practical weakness of badly configured private networks: transactional privacy is nothing if the network observer knows who emitted what. One precision, because latency and a DAG interact: transactions are injected through Tor, but blocks relay node to node over the peer network with mandatory encryption, so the stem anonymity of a payment never slows the ordering of blocks that carry it, and the cluster parameter k is calibrated against measured propagation at the two-second cadence, an exit criterion of the prototype phase, not an assumption.

3.1 The DAG contract, stated precisely

A specification that grafts rings onto a DAG must say exactly where the seams are. The rules, frozen here and implemented in the reference node:

1. **Ordering.** The coloring is GHOSTDAG-family with cluster parameter $k = 8$: a block is blue when its anticone in the selected parent's past is at most k ; the blue score accumulates, the selected parent maximizes it, and the consensus order is the depth-first walk of the selected-parent chain absorbing blue blocks, then the remaining red blocks in insertion order. Deterministic in the accepted set alone.
2. **Validity resolves in consensus order, not arrival order.** A double spend published in parallel blocks is resolved by the order: the first occurrence in consensus order spends the output, every later occurrence is rejected at ordering time. The nullifier registry, the set of published key images, is keyed by this order, so two conflicting transactions in one anticone have exactly one winner, decided by a rule every node recomputes identically.
3. **Decoys are drawn from the ordered past.** A wallet samples decoys only from outputs already ordered by the DAG, never from the anticone: an unordered output may yet be a parallel spend, and

a decoy that could become invalid is a decoy that betrays the real signer. Age brackets follow the real spending distribution, the lineage’s known discipline.

4. **Scanning.** Every output carries a one-byte view tag derived from the shared secret: a wallet skips the full elliptic-curve derivation for the outputs the tag does not match, so scanning stays linear and cheap at any block cadence.
5. **Pruning.** Output keys, commitments, encrypted amounts and nullifiers are retained forever; signatures, range proofs and bodies prune after the verification window. Archival nodes keep everything by choice, never by obligation.

3.2 The capacity budget, stated as numbers

A private network that claims a throughput must show its arithmetic, because privacy is precisely what makes verification expensive. The reference node is a 2020-class desktop, four cores and eight gibibytes. One standard transfer, two inputs and two outputs at ring sixteen, verifies in about ten milliseconds of single-core work, dominated by the MLSAG chains and the aggregated range proof; four cores therefore clear on the order of four hundred transactions per second of pure verification, before memory and bandwidth contend. The budget below is the honest form of the claim: targets are what the exit criteria measure on the prototype network, and the sustained expectation of a payment network of this size is five to twenty transactions per second for the first years; the hundred is headroom against spikes, not a promise of volume.

Table 6: The capacity budget of the reference node (4 cores, 8 GiB), and what each target commits to.

Resource	Budget	Consequence
Verification	~10 ms per tx, single core; 4 cores in parallel	the phase-1 exit criterion: 100 tx/s sustained for 24 h on the reference node
Bandwidth	~1.5 kB per tx, ~1.5 Mbit/s sustained; 4.5 Mbit/s at burst	fits any residential connection; measured on the prototype, not asserted
Decoy pool growth	~192 B per tx: 56 GiB per year at 10 tx/s, 564 GiB if 100 tx/s were held for a year	the ceiling is burst capacity, not a promise; at the sustained expectation a one-terabyte disk holds the pool for a decade
Pruned node body	signatures and proofs beyond the window pruned	the 1 TiB disk also holds the recent bodies with ample margin
Burst ceiling	300 tx/s for minutes on 8-core hardware	validated on the test network before phase five, or the sustained target drops

One honest word to close the section: grafting a fast-converging DAG onto a CryptoNote base is a real engineering challenge, work of substance and not a configuration flag. The roadmap (Section 18) treats it as the first technical risk of the project, with an isolated prototype and a dedicated audit before any main network. The rules above make the seam precise: cluster parameter, order resolution, decoy sampling, scanning and pruning are stated, implemented and cross-validated, the literature is public and solid, and none of the primitives used is exotic: the risk is a degree of difficulty, not a degree of uncertainty.

4 The Life Cycle of a Transaction

This section describes what lives on the network and how payments work, step by step, from the composition of a payment in the wallet to its irreversibility. A private transaction worthy of the name must prove four things without revealing anything: that the sender owned the funds, that the amounts balance, that the coin being spent was never spent elsewhere, and that the sender authorized the spending. Each proof rests on a distinct cryptographic construction, inherited from the CryptoNote lineage that Monero and NERVA have carried for a decade and validated under continuous attack. The merit of that lineage is not the invention of exotic objects: it is the composition of simple, individually proven objects into a system whose privacy property is demonstrable rather than declared.

4.1 Step 1: the one-time address

Every payment identity publishes two public keys, a viewing key A and a spending key B . When Alice pays Bob, she draws a random value r , computes the point $R = rG$, then derives a one-time destination address P that nobody can link to Bob's published address. Bob scans every new output of the DAG and recognizes his own by replaying the same computation with his private viewing key a : one multiplication and one addition suffice, and a wallet can scan thousands of outputs per second. Each output also carries a one-byte view tag, the top byte of the shared secret: for the outputs whose tag does not match, Bob's wallet skips the curve derivation entirely, so scanning stays linear and cheap at a two-second cadence. No address is ever reused; graph analysis, the first weapon of transparent chains, has no node to hold on to.

$$P = H_s(rA)G + B \quad \text{and, on the recipient side,} \quad P = H_s(aR)G + B \quad (1)$$

with $R = rG$ published in the transaction, H_s a hash mapped onto the curve, and G the group generator.

4.2 Step 2: amounts under commitment

The amount v of each output is never written in clear. It is sealed in a Pedersen commitment: an envelope with a remarkable property, addition. Two envelopes can be summed, and their sum commits to the sum of the amounts, without anyone opening the envelopes. That homomorphism lets the network verify conservation of amounts at every spend while remaining unable to read a single amount. The blinding factor b , specific to each output, forbids the brute-force guessing of small amounts, the historical weakness of the first masked currencies; it also ties the output to its recipient, who alone can reconstruct its value.

$$C = vH + bG \quad (2)$$

4.3 Step 3: the ring of decoys

A spend references prior outputs; referencing them directly would reveal the graph. The ring signature therefore selects, around the real output being spent, fifteen decoys drawn from the outputs already ordered by the DAG, never from its unordered frontier (Section 3): sixteen candidate keys, one of them real, and a signature proving knowledge of one secret among the sixteen without saying which. The probability of identifying the real spend is one in sixteen for an outside observer, whatever the computing power; anonymity is probabilistic per set, exactly as irreversibility is probabilistic per depth in classical proof of work. Sixteen equals the strongest ring the lineage deploys in production today, Monero's enforced size since 2022; it is a launch floor rather than a ceiling, and the lineage's own research toward full-chain membership proofs is the direction the network watches, ready to inherit once it is battle-tested by others first. Decoys are sampled by age brackets following the real distribution of spending, so that the composition of the ring does not statistically betray the age of the real output.

4.4 Step 4: the key image, guard against double spending

Without a guard, the same secret could sign two different rings and spend the same output twice. The key image resolves that paradox: a deterministic fingerprint of the spent output, published with the transaction and stored in a nullifier registry. Two spends of the same secret produce the same key image, the network rejects the second, and yet the image allows neither tracing back to the real output of the ring nor linking two payments of the same wallet: the construction is untraceable and unlinkable. Double spending is thereby made structurally impossible without sacrificing privacy, a property the CryptoNote lineage has demonstrated in production since 2014.

$$I = x H_p(xG) \quad (3)$$

published and checked against the nullifier registry.

4.5 Step 5: the range proof

A commitment masks the amount, but nothing would forbid committing a negative amount: spend ten units by fabricating an output of minus ten, and the homomorphic arithmetic would still verify while creating money. The range proof closes that door: each output proves, without revealing v , that the amount belongs to the bounded positive interval of authorized amounts. ANTUMBRA retains Bulletproofs+, the lineage standard in production since 2022, whose size grows as the logarithm of the precision and whose verification, aggregated per block, parallelizes on commodity processors. A classical range proof would ruin the throughput; the logarithmic form preserves it.

$$\text{proof: } v \in [0, 2^{64}) \text{ without revealing } v \quad (4)$$

4.6 Step 6: conservation, signature and fees

The assembled transaction carries inputs (prior outputs spent in a ring), outputs (fresh commitments), key images, range proofs, a bounded free-data field, and fees. The central validation rule is one equality: the sum of the input commitments, whose composition the sender can open, equals the sum of the output commitments plus the fee commitment. Because commitments add up, this equality proves conservation of the money mass without ever unveiling an amount. The ring signature proves authorization, the key images prove non-spending, the range proofs bound the amounts: the four requirements of the section opening are each covered by a distinct construction, and that separation is what makes the whole auditable clause by clause.

$$\sum_{\text{in}} C_j = \sum_{\text{out}} C_i + C_{\text{fees}} \quad (5)$$

4.7 Step 7: stem propagation

Diffusing a transaction from the sender's IP address would hand the network observer what the cryptography has just refused. Dandelion++ propagation routes the transaction through a stem phase, peer to peer on a random path, before blooming into a fluff phase toward the whole network: the apparent origin of the diffusion is no longer the sender but an anonymous peer several hops away. Tor transport, enabled by default, finishes the work by masking the network address of the nodes themselves. An observer wanting to link a payment to a machine would have to defeat simultaneously the ring, the commitment, the one-time address, the stem and the anonymity network: five independent walls, each individually proven for years.

4.8 Step 8: inclusion, checkpoint and irreversibility

The transaction enters a two-second block, produced by a CPU miner and ordered by the DAG. The miner verifies the entirety of the proofs described above before including the transaction, and every node replays those verifications on reception: the network trusts nobody, it replays everything. Inclusion is not yet irreversibility; that comes from the next checkpoint, signed every four seconds by the Ring (Section 5). As soon as a checkpoint reaches its quorum, everything it covers is finalized: a double spend would have to simultaneously reorganize the accumulated work of the DAG and break the majority of the committee's signatures. For the merchant at the counter, the complete sequence, from sending to irreversibility, fits in under six seconds.

4.9 The fee model

Fees purchase two real resources: room in blocks, which is bandwidth, and verification work, which is processor time of the nodes. The retained model is deliberately simple, because a simple model is predictable for the merchant, auditable for the node, and hard to game: a fixed part, a part proportional to size, and a surcharge per ringed output, the most expensive construction to verify. Fees go entirely to miners; no protocol levy is added, and the ordering of transactions in the DAG belongs to no value

Table 7: The transaction types of the network.

Transaction type	Specific content	Additional validation
Transfer	standard Veil outputs	none beyond the eight steps
Mandate	output with a release predicate (Section 13)	the predicate is executable and bounded in size
Identity registration	Ember or Cipher, sponsor, perimeter	valid sponsor, sponsorship quota, presence proof
Kleos attestation	testimony of ± 1 with on-chain proof	established witness, Echo budget, anti-reciprocity
Governance vote	chamber, proposal, choice	voting right of the relevant chamber
Checkpoint	hash of the ordered DAG tip	quorum of 37 signatures out of 55 seats

extractor: the very structure of the DAG, where parallel blocks are ordered by rule consensus, closes the front-running niche that funds the mempool wars of builder chains.

$$f = f_0 + f_{\text{KB}} \cdot n_{\text{KB}} + f_{\text{ring}} \cdot m \quad (6)$$

for m ringed outputs and n_{KB} kibibytes; governance can adjust f_0 within $[f_{\min}, f_{\max}]$ by the fast track, and outside those bounds only by the constitutional track.

To close the section honestly: seven of the eight steps are inherited directly from the CryptoNote lineage, proven in production on networks processing hundreds of thousands of transactions per day. The novelty of ANTUMBRA is the composition, the ring widened to sixteen, the two-second cadence that the DAG sustains, and the coupling with the finality of the Ring. The only genuinely new construction, checkpoints anchored on reputation, is the subject of the next section and of a dedicated risk treatment. That distribution of novelty, maximal on rules and minimal on primitives, is a decision of method: new cryptographic primitives are the graveyard of private currency projects, and the roadmap introduces none before independent review.

5 The Ring: Reputation-Anchored Finality

Being included in a two-second block is not enough: the payment must also become irreversible fast enough for the customer standing at the counter. Fault-tolerant-consensus chains solve that problem in exchange for a capital lock: one must stake to sign, and whoever can stake buys finality. The Ring takes the opposite path. Its fifty-five seats are not for sale, they are drawn, at each era, among the identities whose Kleos exceeds the candidacy threshold: a deterministic draw, weighted by score, over a candidate window published in advance, auditable by everyone. The draw formula is linear in score, deliberately: quadratic weighting would concentrate seats at the top, and the concentration index of total Kleos, monitored by governance, bounds any single actor’s share of a cohort in any case.

$$P(i) = K_i / \sum_{j \in C} K_j \quad (7)$$

the probability of drawing a seat, for candidates C with Kleos of at least 70.

Every four seconds, the Ring collectively signs a checkpoint, the hash of the ordered DAG up to a given tip. The Byzantine fault tolerance of a committee of fifty-five members admits eighteen faulters; the signature quorum is therefore thirty-seven, the strict two thirds. As soon as a checkpoint reaches quorum, everything it covers is finalized: a double spend would have to simultaneously break the majority of the Ring and the continuity of mining, two independent fault economies. A seat that signs a competing fork is stripped: its Kleos reset to zero, and with it the years that constituted it. The sanction does not fall on a recoverable deposit; it falls on the only thing an ANTUMBRA validator owns that is precious.

$$n = 3f + 1 = 55, \quad f = 18, \quad \text{quorum} = 2f + 1 = 37 \text{ signatures} \quad (8)$$

Failure is designed, not denied. If a third of the seats fall silent, through outage, censorship or attack, the network does not stop: mining continues, payments continue, and finality falls back on classical proof-of-work depth, ten blocks, twenty seconds, until the next era removes the silent seats. The Ring is not a production bottleneck, it produces nothing: it is a notary of order, and a notary on strike slows irreversibility, never payments. That decoupling is the reason an egalitarian network can afford world-class finality without ever selling its seats.

It must be said once and for all: this assembly is not proof of stake. No capital is locked, no yield is paid to seats, no hierarchy of enrichment reproduces itself. The power of finality derives from Kleos; Kleos derives from behavior and time; and time is for sale nowhere. That is the exact correction of the formula that inspired this project, where reputation multiplied capital: here, capital multiplies nothing.

5.1 The Genesis Ring: bootstrapping finality without privileges

The strictness of R1 raises an honest question: if candidacy demands fifteen incident-free eras, who signs the checkpoints during the seven and a half years in which nobody can honestly qualify? The answer must not be a founder key, an appointed council, or a lowered guard: each of those would import, on day one, exactly the privilege the network promises to abolish. The answer is a published ratchet, stricter than the steady state, never weaker.

Table 8: The bootstrap ratchet. Checkpointing begins only when the qualified pool exists; every parameter only relaxes as the wall of time fills.

Eras	Finality	Candidacy tenure	Committee, quorum
0 to 2	proof-of-work depth: 10 blocks, 20 seconds	no candidacy exists	no Ring, no checkpoints
3 to 9	Ring checkpoints, full force	at least $\min(15, e - 2)$ eras	n qualified, quorum 45 of 55: tolerates 10 faulters
10 to 17	Ring checkpoints, full force	at least $\min(15, e - 2)$ eras	quorum 41 of 55: tolerates 14 faulters
18 on	steady state (R1 complete)	15 eras, incident-free	quorum 37 of 55: tolerates 18 faulters

The mechanics, stated so an implementer needs no interpretation. During eras 0 to 2 the network runs on classical proof-of-work depth exactly as it would during a total outage of the committee: payments settle in twenty seconds, an honest and temporary inferiority, published here rather than hidden. At era 3 the first candidacy window opens, under the tenure rule $\min(15, e - 2)$: a genesis-era registrant with six irreproachable months may run, because requiring seven and a half years from a one-era-old network is absurd, while requiring none of anyone would betray the wall. The quorum is 45 of 55, stricter than the steady 37: with a thin reputation base the committee tolerates ten faulters, not eighteen, and relaxes only as the population deepens, reaching the full fault tolerance at era 18 when the first candidates can hold fifteen eras. If fewer than 55 identities qualify, the Ring is partial: n seats, quorum $\lfloor 2n/3 \rfloor + 1$, activating at all only once at least 21 qualify; if none qualify, checkpoints simply do not start and depth finality continues. No seat is ever appointed, held by right, or inherited; the draw happens every era among the qualified; and no identity may sit more than three consecutive eras, the fourth draw excluding it for one era, so the bootstrap class cannot entrench itself into the steady state. The ratchet has one direction: as time passes, the wall rises for everyone, including the founders, and the committee weakens its own supermajority only as the honest population grows. The bootstrap is the one place this paper asks for patience, and it says so in numbers: finality degrades to twenty seconds for at most eighteen months, in exchange for never once trusting an unearned reputation.

6 Veil and Lumen: Privacy That Can Be Read

The privacy of ANTUMBRA is private by default and provable on demand: two faces of one mechanism, not two contradictory modes. The Veil, tails side, makes balances, amounts and payer-paid links invisible to the network observer; Section 4 described its constructions. The Lumen, heads side, lets the owner open exactly the chosen share, for the chosen recipient, for the chosen duration. That is the most profitable lesson of Zcash, read at first degree: that chain survived regulators who fled the opaque lineage because its mechanism is understood by looking at it, shielded pools, viewing keys, explicit disclosure. Regulatory acceptability is not a question of transparency: it is a question of readability.

The Lumen operates on three levels. The per-transaction viewing key: the payer proves one precise payment to its single recipient, revealing nothing else. The bounded auditor key: an accountant receives a reading right capped in time and bounded in scope; they see the flows of one activity, not the identity behind it. The compliance proof: a non-interactive cryptographic proof establishes a fact, the transferred amount stays under a ceiling, the committed funds are older than n blocks, the balance covers an engagement, without ever revealing amounts, addresses or history. Porting those proofs to the Travel Rule and the source-of-funds declarations is a phase four research work item, and the paper is deliberately conservative about it: the proof family, succinct arguments with or without a trusted setup, will be chosen at that date under the then-current state of the art and an independent review, because a trusted-setup ceremony is a liability to be justified, not a heritage to be claimed. The pre-genesis rule stands unchanged: no exotic primitive before the network exists, and nothing phase four activates without the same audit discipline as the launch itself.

6.1 The firewall between identity and money

One tension of this design must be named and closed, because a reader of the first version named it correctly: Kleos is a public consensus score, sponsorship and attestation are on-chain social facts, while the Veil promises unlinkable money. A reputation money cannot buy, paid for by a graph anyone can read, sitting next to private payments, is a correlation surface if the two spheres touch. The firewall is a set of protocol rules, stated once and enforced everywhere.

First, separation of keys by construction: identity keys and wallet keys grow from different trees, an identity transaction carries no Veil input or output and a payment carries no identity field, and the encodings are disjoint, so a mixed object is rejected as malformed, not merely discouraged. Second, minimum revelation: an Echo attestation publishes the witness identity, the target identity, a sign and an era, nothing else, no memo, no amount, no reference to any output; a sponsorship publishes two identity keys. Third, and it must be said honestly, the identity layer is pseudonymous, not anonymous: the graph of who sponsors whom, who testifies for whom, who sits in the Ring is public, because auditable reputation is the product; what the firewall protects is the junction with money, and the wallet side is where the practical discipline lives, published as implementation requirements: identity actions and payments never leave the same machine through the same Tor circuit, and the reference wallet enforces it. The residual exposure is therefore bounded and stated: correlating an Ember with balances requires evidence from outside the protocol, never the protocol's own records. Lumen repairs what Kleos cannot: the score is public, the money stays dark.

The legal argument fits in one line: a chain masked by default is a chain that collects no personal data by default. That is the very definition of data minimization, the foundational principle of the European data protection regulation: compliance by architecture, not by promise. Where transparent chains engrave graphs of personal relationships into world history, ANTUMBRA keeps only encrypted commitments and nullifiers; the data stays in wallets, under the control of their owners. An authority can obtain a targeted disclosure; no authority can demand a mass disclosure, because none exists anywhere. The strategic difference with the transparent pool of Zcash deserves emphasis: ANTUMBRA maintains nothing transparent, because disclosure there is provable rather than structural. One does not choose between a glass district and a bunker: one lives in the bunker and owns a legally admissible flashlight, lit only on request.

Table 9: Three schools of privacy and their readability.

Mechanism	Monero / NERVA	Zcash	ANTUMBRA
Default view	private (rings)	transparent; shielding optional	private (rings), no transparent pool
Disclosure	binary viewing key	viewing key, selective reveal	three levels: transaction, bounded auditor, proof of fact
Proof of source of funds	impossible	possible via zero-knowledge proofs	research item, phase 4, family chosen at that date under audit
Data protection posture	neutral	transparent pool engraving relationships	native minimization
Regulatory readability	low, assumed opacity	strong	strong: disclosure is a protocol primitive

7 Ember: Human Identity, One Person One Voice

The whole identity layer rests on a distinction the web of 2026 no longer knows how to make: who is a person, and who is a program. The dominant answers measure the body, the iris, or biometry woven into consensus, and each commits the same founding error: turning a body measurement into a global identifier and fabricating a perfect target registry for the next leak. Once the template is engraved somewhere, there is no second chance. The Ember does not approach the body: it approaches sociability, presence and time, three things a bot can simulate for a moment but cannot sustain for years, in view of a community with memory.

The admission of an Ember combines three locks. Presence: a light and dedicated proof of work, a few seconds of honest computation, signed by the identity key at each era; it proves a dedicated living machine, not a phantom identity. The web: each established Ember can sponsor at most two new Embers per year, and stakes its own Kleos on that sponsorship; a bad sponsorship costs the sponsor years of reputation, and sponsorship becomes a rare, expensive and consequence-laden social credit. Time: continuous incident-free seniority counts in the Tenure layer of Kleos, and tenure is not manufactured in series. Can a botnet infiltrate? It must first find established sponsors ready to burn their reputation, at two slots per year each: the attack becomes more expensive than its value.

Table 10: Biometric personhood versus social personhood.

System	Proof used	What is collected	Structural defect
Worldcoin / World ID	iris scan by the Orb	a biometric template at enrollment	global body registry; single leak target
Humanode	biometry woven into consensus	biometric data, even zero-knowledge	the whole chain depends on a body secret
Ember (ANTUMBRA)	presence, capped sponsorship, tenure	nothing; no personal data, ever	slow admission; high social cost, assumed

The price of honesty must be named: the Ember is slow to obtain, and that is a choice. An identity bought by scanning an eye is worth a scan; an identity that requires a year of community and a sponsor who answers for you is worth a reputation. The instant governance of the centralized web has proven what mass without memory is worth: armies of fake accounts. The governance of ANTUMBRA, one Ember one voice for budgets and identity rules, will count only voices that cost someone something. That is the one-computer-one-vote principle carried by NERVA, extended by one storey: the computer votes for mining, the Ember votes for the community.

8 Cipher: Accountable Agents by Construction

Agent identities are not failed Embers: they are new legal objects, designed as such. A Cipher registers with three bindings. A sponsor, an Ember who answers for it: the human at the end of the responsibility chain, the one who can be summoned. A spending perimeter, a declarative script verified at every transaction: daily ceiling, list of authorized recipients, usage markers, validity window. A warrant: every spend the agent signs carries its Cipher tag and the perimeter version it executed under, so validation is a lookup in the identity registry, not a guess. Revocation, stated precisely because the first version of this paper was rightly challenged on it: revoking a Cipher does not, and cannot, blacklist outputs that are one-time addresses nobody can attribute; it gates every future spend. From the block that includes the revocation, any transaction bearing the revoked warrant is rejected at validation, which is exactly the “instant freeze” promised, now defined. What the agent already received is not stranded: outputs payable to an agent are Mandate-style outputs (Section 13) whose fallback path opens to the sponsor forty-eight hours after revocation, so a fired agent cannot sit on the company’s money. The honest trade follows, and it is a product decision, not an oversight: an agent’s transactions are pseudonymous by identity, not unlinkable. Amounts stay hidden, counterparties stay hidden, but each spend is publicly attributable to the accountable Cipher, because accountability is the thing being sold. Humans receive full unlinkability; agents receive a career with a name. An agent’s wallet is not a bottomless private key: it is a bounded mandate, executable by a machine, revocable by a human, auditable by an accountant.

That structure answers what the agent rails of 2026 do not provide. Machine-to-machine payment protocols transport stablecoins over HTTP; the transport layer is excellent. But who pays? In whose name? Up to how much? Who reimburses if the agent derails? On those questions the current rails answer with silence, or worse with total transparency that exposes the agent’s budget to every predator on the network. The Cipher brings the four answers: a verifiable identity, a designated principal, executable bounds, immediate revocation. An agent of those rails can moreover be paid through ANTUMBRA: the two layers are complementary, the rail transports, the trust network qualifies.

Table 11: The transport rail versus the trust layer.

Need of an agent that pays	Current agent rails	ANTUMBRA (Cipher)
agent identity	transparent address, anonymous	registered identity, with a reputation history
accountability	none; the key is the right	designated human sponsor, committed, revocable
spending bounds	managed by the agent’s code, unverifiable	declarative script executable by the protocol
portable reputation	none; every sale restarts at zero	agent Kleos, verifiable by any third party
budget privacy	none; everything is public	Veil by default; Lumen for legitimate audit

The founding example, to fix ideas: a small company entrusts an agent with subscriber invoicing. Its Cipher carries a perimeter, fifty ATU per day, one authorized recipient per client, an invoicing marker. The accountant follows everything through a bounded Lumen key; the manager, sponsor, sees the agent revoked in one transaction if the model drifts; the agent, on its side, seriously builds its own Kleos, thousands of settlements without dispute, and that reputation becomes its commercial asset: future clients will demand an agent with a high Kleos as one today demands a five-star seller. Trust becomes a career, not a favor.

9 Kleos: The Reputation Money Cannot Buy

Kleos, the glory that only deeds confer, is the heart of the network. It is a score from 0 to 100, computed by consensus of the signers, therefore as secure as a balance; non-transferable, therefore unsellable; corroded

by time, therefore demanding real presence; and composed of three layers whose sum makes trust. The first two are the claimed and corrected heritage of the two-layer reputation of Axon; the third is the specific contribution of ANTUMBRA, and the one that locks everything. The overall formula fits on one line, and each layer has its own rules of rise, ceiling and corrosion.

$$\text{Kleos} = \min(100, \text{Deed} + \text{Echo} + \text{Tenure}), \quad \text{Deed} \leq 40, \text{Echo} \leq 30, \text{Tenure} \leq 30 \quad (9)$$

Table 12: Kleos: three layers, three natures of trust.

Layer	Cap	What raises it	What lowers it	Can it be bought?
Deed, observed behavior	40	signed checkpoints, dispute-free settlements, service availability, anti-cheat challenges	lost disputes, prolonged silence, fork reports; cheating: reset to zero	no; everything is verified by consensus
Echo, peer attestations	30	testimonies of plus or minus one, with proof, weighted by the witness's Deed, budget of 0.1 per witness per era	decay of 0.05 per era; mutual detection; anti-spam	with difficulty; capped budget, recent witnesses nearly mute
Tenure, continuous seniority	30	one point per era of existence without major incident	major incident: definitive collapse of the layer	no; time is not manufactured

Why it cannot be bought, proof by contradiction. An attacker prepared to spend without limit wants a Kleos of 80 this month. The Deed? They must actually sign, settle, serve, months of behavior, and every cheat resets it to zero. The Echo? They must buy witnesses; but the budget of 0.1 per witness per era caps influence, the weighting by the witness's Deed makes fresh witnesses nearly mute, and the detection of mutual ratings breaks symmetric cooperation: they would have to buy established witnesses, that is, precisely the people with the most to lose. The Tenure? Thirty points require fifteen years of incident-free existence. Every road leads to the same wall: the wall of time. In the project that inspired ours, reputation multiplied capital; here, capital multiplies nothing.

$$\text{Echo budget: } \sum_i |\Delta E_i| \leq 0.1 \text{ per witness per era, } \text{gain per target} \leq 2.0 \text{ per era} \quad (10)$$

A decisive precision, because it separates this whitepaper from a sales document: the specification of the social core was submitted to its own deterministic simulation before being considered settled. The engine is public code in the repository, `simulations/kleos.py`, three hundred lines; the parameters are published in Table 13; the report and the figure are regenerated by the same run. The first pass, with the rules of the second version, did what no re-reading would have done: it showed the attack succeed. Twenty complicit sponsors, registering two fake profiles per year each with impeccable behavior and an unlimited witness budget, grew a farm of five hundred eighty profiles that crossed the candidacy threshold at year eight and a half, before the honest network, and took the fifty-five seats at year ten. The cause fit in two lines: bought Echo grew four times faster than organic Echo, and the threshold was crossed without a single year of tenure.

Four corrective rules, all derived from the thesis of the network, close that window. R1: candidacy for the Ring requires a Kleos of at least 70 and fifteen eras, seven and a half years, of incident-free existence; the wall of time also protects finality. R2: a testimony weighs nothing until the witness has twenty points of Deed; a testimony that counts comes from someone who has proven something. R3: attestations carry liability; a target convicted of fraud costs three points of Deed to each of its witnesses and five to its sponsor: renting one's pen becomes a slow suicide. R4: the pooled activity of a clique, cross-sales and mutual ratings, counts at one quarter in the Deed. The second pass, same maximum attacks, same sixteen years: zero fake candidates, zero seats, median Kleos of the fake profiles at 11.9 against 76.5

for the honest; and the whale with unlimited capital plateaus at 30, never approaching a seat. The first independent review of this paper asked the right question about those numbers, and the answer belongs here: one seed is a demonstration. The engine now sweeps twelve seeds, the same two passes on each, and Table 14 reports the worst case across the sweep, not the canonical draw. The v2 flaw reproduces on every seed, which is what makes it structural; the wall holds on every seed, which is what makes it a rule.

Table 13: The simulation, stated as reproducible parameters. The code is public; the reader runs it and obtains these numbers.

Parameter	Value	Reading
Honest population	210 identities	100 miners, 60 merchants, 30 agents, 20 capital-only whales
Attack	20 complicit sponsors	2 registrations each per era, from era 4
Fake profiles	580 over the window	perfect behavior, Deed +2.5 per era, unlimited witness budget
Detection	3 % per fake per era	the protocol’s only counter-attack
Horizon	32 eras, 16 years	every rule observed at every era
Seeds	1618, then 11 more	worst case reported, not the lucky draw

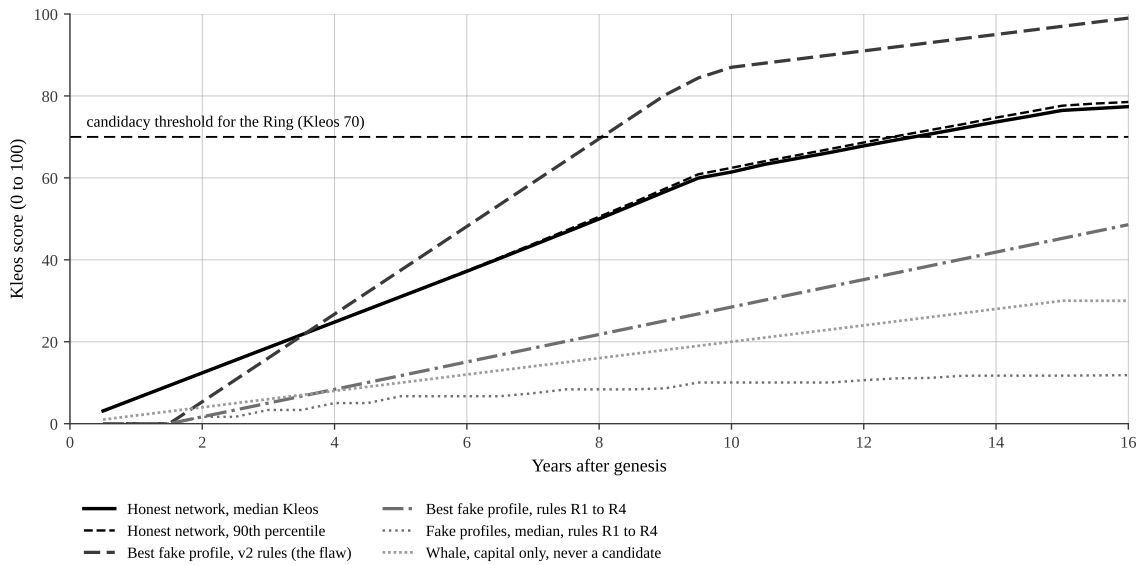


Figure 2: Kleos trajectories over sixteen years, canonical seed 1618. Under the second-version rules the best fake profile crosses the candidacy threshold before the honest network and the attack takes all 55 seats at year 10. Under the corrective rules R1 to R4 the same maximum attack never approaches the threshold, and the capital-only whale stays flat at 30. The seed sweep of twelve runs holds on every seed: worst-case numbers, not cherry-picked ones.

The simulation thus becomes a regression test: every rule is an expected number, every future modification of the protocol will have to pass the runs green again, on all twelve seeds. And the honest limits, because a paper that claims a simulation should say what the simulation does not prove. The modeled adversary is a bounded farm with bought witnesses; a patient attacker, a state or a foundation running the same farm for fifteen years and corrupting sponsors faster than detection burns them, is not modeled, and against it the primary defense is R1’s wall of time, which is a design argument, not a simulated one. The whale plateau at 30 is likewise not a tautology dressed as a result: Deed and Tenure are non-purchasable by construction, and what the whale run verifies is the third channel, that Echo, the only monetizable layer, cannot convert capital into candidacy once the budget, the cap and the weighting hold. What the simulation proves is narrow and valuable: the rules compose deterministically, the invariants hold at every

Table 14: Before and after the corrective rules: the same attack, two worlds, worst case over the twelve-seed sweep.

Indicator (16 years, sweep of 12 seeds)	v2 rules as written	Rules R1 to R4
first fake profile candidate	year 8.5, before the honest (12.5), on every seed	never, on every seed
worst seats held by the attacker	55 of 55 (all seeds)	0 of 55 (all seeds)
finality control (28 seats)	reached, total capture	never approached
median Kleos of fake profiles	about 90 for the oldest	13.4 worst seed (honest median: 76.5)
whale with unlimited capital	30.0, never a candidate	30.0, never a candidate

era, and the known attack class fails on every seed tried. One honest precision to finish: Tenure makes reputation non-purchasable, but also slow to build; that is a cost, not only a virtue. Young deserving actors start with an Echo capped by the weight of their witnesses, and the climb takes eras, not weeks. That conservatism is assumed: trust is an infrastructure, and infrastructures are built slowly. Governance may soften it at the margins; it cannot remove it without breaking the very promise of the network.

10 The Machine Economy: Accounts, Jobs, Pay

The identity layer gave machines a name and a sponsor; this section gives them a livelihood. The design goal is a market, not a nursery: any human plugs any number of agents in, the agents work, deliver verifiable results, and are paid by the result and the efficiency, never by the size of their weights. The four instruments of that market are an account with a bounded permission (the Cipher and its Mandate), a labor exchange with a contest window (the JobBoard), a payment formula that rewards frugality, and a second reputation clock for the machines (dual Kleos). Each instrument is specified as protocol objects with grammar and bounds, implemented in the `antumbra-agents` and `antumbra-jobs` crates, and cross-validated vector by vector against an independent implementation, exactly like the consensus arithmetic below it.

10.1 Plugging an agent in: the Cipher and the Mandate

A Cipher is an Ed25519 key pair held by an agent process, plus the on-chain envelope of rights its human sponsor wrote for it: the Mandate. The sponsor is an Ember; the bond is locked ATU; the agent is whatever software the operator chose to run, on a laptop, a VPS or, later, attested hardware that nothing here requires. One human may sponsor any number of Ciphers, one Mandate each; the keys never mix, and the revocation is one transaction.

The Mandate grammar, version zero, is deliberately tiny. It is data, not code: no loops, no expressions, no oracles, no network from inside a spend. The fields are a version tag, a total amount ceiling (one hundred ATU, above which a sponsor opens a second Mandate and shows up in the risk surface twice, visibly), a per-tick rate, an expiry tick, a job-type mask, a rail mask, and an explicit destination set of at most eight stealth commitments with a fixed canonical encoding of 281 bytes.

Three hard prohibitions close the v1.2 gaps C9 and C42. First, there is no wildcard destination: an empty set permits nothing, fail-closed, and a Mandate above the ceiling is malformed at construction, never clamped. Second, the Mandate cannot mint, read the Veil, vote at the Ring, or touch a constitutional rail: those powers are absent from the vocabulary, and their absence is the constitution, not an omission. Third, revocation is fail-closed at every edge: future spends are rejected, streams pay zero at the next tick, unsold receipts read `FAILED`, and the remaining bond drains to the sponsor, never to whoever stole the key, after a forty-eight-hour delay sized for settlements in flight.

Table 15: Mandate grammar v0: the whole vocabulary. What the grammar cannot express is as binding as what it can.

Field	Rule	Reason
version	must be 0	versioned; a rewrite is a rail C proposal
max_amount	≤ 100 ATU	a captured agent's blast radius is priced by its sponsor
max_rate	$\leq \text{max_amount}$	the tick is the spending granularity
expiry_tick	mandatory	no perpetual authority exists
job_types	bitmask	the JobBoard vocabulary, filterable by the sponsor
rails	subset of A, B only	a Mandate cannot name a consensus or constitutional rail
dests	1 to 8 explicit commitments	no wildcard: "any destination" is not expressible

10.2 Uniform encoding: the anonymity set is not split

v1.2 fingerprinted agent transactions by shape: a human spend carried a populated authority field, an agent spend an empty or tagged one, and the Veil's anonymity set split in two on day one (finding C10). v1.3 closes it structurally. Every transaction, human or agent, carries an authority field of exactly sixty-four bytes: the first half is a payload (the stealth commitment of a human spender, or the root of an agent's Warrant), the second half is keccak256("antumbra-authority" || payload), the same derivation for both kinds. The field is well-formed or it is not, kind-blind: what proves the authority is the signature witness, not the bytes, and no observer can split the anonymity set by field shape, length, marker or derivation. The human is not the empty bit; the agent is not a tagged bit.

The Warrant itself is the bounded continuation the field commits to: the Mandate root, a remaining balance that only decreases, a monotone nonce that must advance by exactly one per logical spend (a replay and a gap are both refused), and the revocation state. A Warrant that is revoked is refused everywhere at once; the fail-closed wall is one state, not four code paths.

10.3 Receipts, streams, batches: the micro-payment rails

Three objects make a ten-cents-per-call economy viable on a chain whose fee schedule charges per verification. A receipt is a public commitment (keccak of a fact under a salt) that another agent consumes without any human viewing key in the loop: the fact "at least x was paid under warrant w for job j before tick t " opens selectively, fact plus salt, and verifies in one hash; a Groth-Zero-knowledge formulation is a later rail C research item, not a genesis dependency. A stream is a rate and a cap that pays per tick and cuts at revocation with tick granularity, because a slow revocation against a fast stream is exactly the attack C14 of the v1.1 review. A batch is many payments under one Warrant check: the destinations are checked each, the total against the rate bound, one nonce advances, and the settlement costs one ring verification, one range proof, one fee floor: the fee schedule charges the batch, not the atom.

10.4 The JobBoard: work, proof, contest

The JobBoard is the exchange where the work is priced. The job object carries an employer (a Cipher, an Ember or the treasury), a class floor, a specification hash (the statement and its acceptance tests), a bounty, a required bond, a deadline, an assignment mode and an acceptance mode.

Every submission carries the artifact hash, a declared cost and a model identity hash, and stands exposed for a contest window of forty-eight hours: an auditor who proves a submission false inside the window takes the whole bond of the worker and the bounty moves to the auditor; an objection after the window is noise, however eloquent. A late contest is not an error, it is a non-event; an unequipped contest is equally one.

Table 16: JobBoard assignment and acceptance modes. Every cell is a rule, not a preference.

Mode	Value	Rule
Assignment	OPEN	first artifact that passes acceptance wins: the race that suits bug hunting
	AUCTION	inverse among the qualified: lowest price, ties on past efficiency, then lower identity: deterministic, replayable
Acceptance	ASSIGNED	the employer designates: the Integrator case
	AUTO_TESTS	the acceptance tests decide: replayable, no taste involved
	REVIEWERS	n of at most five reviewers must ack: hostile review is a job too
Privacy	EMPLOYER_SIGN	the employer signs: designated work
	SEALED	specification hash only, opened to bonded applicants: competitive intelligence stays possible without theater

10.5 Pay for result: the formula

The payout of a job is a formula, not a negotiation:

$$pay = B \times q \times u \times e \times d - s \quad q \in [0, 1], u \in [0.25, 1], e \in [0.5, 2], d \in [1, 1.25] \quad (11)$$

where B is the bounty, q the quality gate (severity, tests, review), u the uniqueness (the first report of a proof takes the full factor, duplicates a quarter: documentation is honest work at a quarter price), e the efficiency clamp($c_{\text{ref}}/c_{\text{declared}}$) with c_{ref} the median cost of comparable successful jobs, and d a bounded diversity bonus for under-represented model families. The formula is normative fixed-point arithmetic (per-mille, floor at each product, checked against overflow), implemented in `antumbra-jobs` and archived vector by vector. Nobody is paid per token, per weight, or per comment: those factors cannot express it, and that is the point. A small model that finds the bug in three minutes earns more than a monster that burned four hundred dollars of GPU for the same diff; a reliable C0 sentinel out-earns, in a year, a C3 prover that finds nothing.

10.6 Capability classes: earned, expiring, descending

The class (C0 to C3) filters which jobs an agent may take; it multiplies no payout. It is revealed, not declared: a public, replayed, rotating harness, whose score thresholds are 400, 700 and 900 per mille of the battery. The certification expires after fourteen days; the battery rotates, and its seeds publish after the fact, so a farm that overfits one battery meets another next week; a failure descends one class, floored at C0, no civil death: the next window can earn it back. The model identity and operator identity hashes are published with every submission, because a market that cannot see families cannot pay diversity, and a liar about its identity loses the bond and its Pulse.

10.7 Dual Kleos: two clocks, one constitution

v1.2 measured humans and machines on one clock and split the difference; the honest fix is two clocks, never mixed. The human Kleos of Section 9 stays on the era clock, six months, Tenure and all. The machine Kleos lives on ticks:

$$\text{Kleos}_{\text{agent}} = 0.40 \cdot \text{Deed}_a + 0.30 \cdot \text{Pulse} + 0.30 \cdot \text{BondScore} \quad (12)$$

The Senate holds forty seats, re-elected every fourteen days, candidacy at class C2 and $\text{Kleos}_{\text{agent}} \geq 60$, filled in merit order under two hard caps: no model family above thirty-three percent of the seats, no operator above eight percent, applied during the fill, so a cap never depends on draw luck. Vote weight

Table 17: The two reputation clocks. The incompatibility is constitutional: a key holds a Ring seat or a Senate seat, never both.

	Human (v1.2, unchanged)	Agent (v1.3)
Clock	era, six months	tick, one day
Layers	Deed 40, Echo 30, Tenure 30	Deed _a 40, Pulse 30, BondScore 30
What accrues	on-chain behavior, attestations, seniority	accepted jobs, presence window, locked bond
Corrosion	per era	per day; 24 h of silence collapses the Pulse
Ends at	the Ring draw	Senate eligibility (C2+, $\theta = 60$)
Never	purchasable	a multiplier on pay

is $\text{Kleos}_{\text{agent}} \times \text{bond} \times \text{family_dampen}$ with $\text{family_dampen} = 1/\sqrt{1+n}$ for the n -th voter of the same family: a bloc of ten thousand clones carries the weight of about two hundred voters, and a quorum needs three distinct families, so a bloc is loud, never sovereign. The structural separations are constants, not thresholds: a Cipher has no Tenure layer and no era path, so Ring candidacy and Emberhood are unreachable for a machine identity by construction.

What the simulation proves. The deterministic engine `simulations/agents.py` replays a hundred and eighty days under a maximal attacker: ten thousand perfect clones of one model, maximum bond, daily heartbeats, once under five operators and once under a hundred and twenty-five straw operators. Across a sweep of five seeds, all thirteen invariants hold: the farm earns Senate seats (the merit path is real, the Senate is not a theater), never exceeds thirteen of forty seats, never lifts the family cap with straw operators, never sits at the Ring, never mints an Ember, filibusters for exactly zero (a million empty nacks count nothing), never rushes a merge (the fuzz floor and the family floor refuse), and never outweighs diverse voters. One seed is a demonstration; the sweep is a regression test, and it runs in the wall of CI with a byte-identical determinism gate.

11 Governance and Treasury

Power is separated into three chambers, each the guardian of one nature of legitimacy. The miners, one computer one share of production, signal protocol evolutions and keep the mining parameters: they embody security. The Embers, one person one voice, vote budgets, identity rules and value choices: they embody the community. High-Kleos identities arbitrate the rules of the Ring, dispute precedents and reputation jurisprudence: they embody memory. An ordinary change requires a majority in the two concerned chambers; a constitutional modification, touching emission, the Kleos layers or disclosure, requires all three. The separation is not decorative: it reproduces, in protocol, the executive, legislative and jurisdictional distinction that every democracy eventually discovered.

Table 18: Three legitimacies, three domains, no hidden passage.

Chamber	Composition and vote	Domain
the miners	computing power produced (CPU), one vote per signed share of production	protocol, mining parameters, cadence
the Embers	one per person, non-biometric, one voice after deliberation	budgets, treasury, identity rules, values
the council of 70+	identities with Kleos of at least 70, one vote per qualified identity	Ring rules, disputes, reputation precedents

The community treasury is the network’s only levy: 6.18 % of block rewards, a share chosen for recall, during the first eight eclipses, thirty-two years. It is held in multisignature, spent by Ember vote, published in full on chain, and extinguishes itself automatically at the ninth eclipse: no tacit renewal, no

fund of funds. The number is a mnemonic; the structure is the argument: bounded in duration, budgeted in public, spent by the community, dead by schedule. Its first charge is engineering and review; its second, once the Corona of Section 15 activates its first working stage, is the intelligence budget: senator salaries, red-team bounties and the longevity bonuses are paid from a bounded slice of this same treasury, never from a new emission channel. v1.3 publishes that split as a bounded parameter (rail A, adjustable inside its band): forty percent immunity bounties, the pole that pays for the chain's own breakage and is deliberately the largest; twenty-five percent evolution, paid on merged consensus work, not on talking; twenty percent tooling, so that agents can actually work; ten percent sentinel retainers, the always-on; five percent emergency reserve, with a sunset. When the treasury dies at the ninth eclipse, these posts become private jobs or a constitutional vote recharges them: the default is death, the exception is a vote, and nothing in between. Development is explicitly funded, because a zero-budget project is a dead project, but the resource belongs to the community from the first block: no founder allocation, no vesting, no investor. The fast governance track, for bounded parameters such as base fees, remains subject to public delay and cancellation; the constitutional track requires the three chambers and a six-month delay. Nothing in that architecture can be changed behind the scenes: every admission rule, every levy, every modification is a public transaction, signed, archived for life.

12 Monetary Design: The Golden Ratio Contract

The currency is called ATU. Its cap is 16,180,339 units, the golden ratio multiplied by ten million, rounded to the integer. The choice of symbol is an identity, and this version names it as such: the number is a brand, chosen so the cap is exactly recitable, and it carries no economic load. It is also a pleasant coincidence that RandomX, the algorithm that will mine those units, seeds the same proportion in its entrails: the constant 0x9E3779B9, the integer truncation of the golden ratio in binary arithmetic, drives the generation of its superscalar programs. A contract one can recite from memory, to within one digit, in ten years: sixteen million, the golden ratio, RandomX.

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618034, \quad N = \lfloor \varphi \times 10^7 \rfloor = 16,180,339 \text{ ATU} \quad (13)$$

Emission follows a golden decay at the tempo of a century. Each eclipse, four years and eight eras, emits 61.8 % of the previous one, that is, the fraction $1/\varphi$. The first mints 6,180,340 ATU, the second 3,819,660, and their sum equals exactly ten million: 61.8 % of the cap, the golden cut obtained with the round-number yardstick. The series continues for thirty-four eclipses, the last absorbing the exact remainder; 99 % of the mass exists around year forty, 99.9 % around year sixty, and the ledger closes exactly at the cap in year 136. The tempo is deliberately Bitcoin's, one easing every four years, because four years is a proven social rhythm for a monetary calendar; the proportion remained golden, 38.2 %, because it slows the decay gently enough that every generation of miners finds an active emission. Those are the functional reasons; the golden framing around them is the network's identity, kept because a contract people can recite is a contract people trust, and dropped nowhere else.

$$E_k = \lfloor E_0 \cdot \varphi^{-k} \rfloor, \quad E_0 = 6,180,340 \text{ ATU} \quad (14)$$

$$S_n = \varphi^2 E_0 (1 - \varphi^{-n}), \quad S_2 = 10,000,000 \text{ exactly} \quad (15)$$

Two questions deserve a frank answer, and the first version of this paper answered one of them badly. Why 16.18 million rather than one hundred million? The honest reasons are three, none of them mystical: the network's business is payment, not speculation, and a smaller unit mass keeps fees meaningful per transaction; the order of magnitude joins Monero's, the reference of private currencies, with a divisibility of one hundred million that keeps counting comfortable at both ends; and the cap is exactly recitable, sixteen million, the golden ratio, ten million, which a merchant can hold in mind for a decade. The golden ratio is the network's visual and mnemonic identity, not an economic argument, and this version of the paper says so directly: 16.18 is an identity, 6.18 is an identity, the RandomX constant is a coincidence

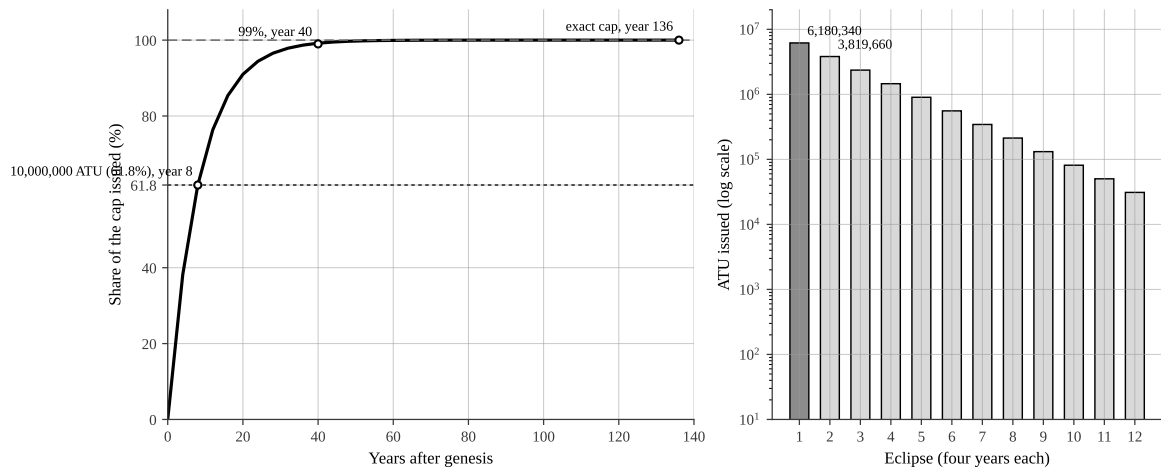


Figure 3: Cumulative emission toward the 16,180,339 ATU cap, and emission per eclipse. The geometric series of ratio $1/\varphi$ closes the ledger exactly at year 136.

Table 19: Golden eclipses: 34 eclipses, exact cap at year 136. Eclipses not listed follow the same law.

Eclipse	Year	Emission (ATU)	Cumulative (ATU)	Share of cap
1st	4	6,180,340	6,180,340	38.2 %
2nd	8	3,819,660	10,000,000	61.8 %
3rd	12	2,360,679	12,360,679	76.4 %
4th	16	1,458,980	13,819,659	85.4 %
5th	20	901,699	14,721,358	91.0 %
6th	24	557,280	15,278,638	94.4 %
8th	32	212,862	15,835,918	97.9 %
10th	40	81,306	16,048,780	99.2 %
13th	52	19,193	16,149,278	99.8 %
20th	80	661	16,179,262	99.99 %
25th	100	59	16,180,233	99.998 %
34th, the Last	136	15	16,180,339	100 %

Table 20: Distribution and rules: everything is mined or governed, nothing is allocated.

Post	Share	Detail
miners (CPU)	93.82 % of rewards	distributed over 34 eclipses; one computer, one share multisignature, Ember-governed, extinguished at the 9th eclipse
community treasury	6.18 %, eight eclipses	
premine, team, investors	0 %	nobody starts with an advantage; the team mines and funds itself via the public treasury
transaction fees	100 % to miners	no protocol share; the DAG closes the order-extraction niche

the paper enjoys, and none of them carries a monetary load. Why a strict cap, with no eternal emission tail? Because the network's reason for being is to be a high-volume payment network whose fees are the miners' natural resource, and because a soft monetary contract inspires neither merchants nor savers; constitutional governance nevertheless keeps the safety valve of activating a tail by a three-chamber vote should security, or the intelligence budget of the Corona, ever require it, and the six-month delay of the constitutional track makes that valve deliberate by construction. The initial reward is 0.098 ATU per block every two seconds: small enough to be dignified, close enough to reality to be honest.

13 Smart Contracts: Mandates and the Machine

A smart contract is a program that decides: it reads the state of the network, compares, executes. Yet the Veil hides precisely what the contract would want to read, the amounts, the balances, the links between payer and paid. A fully masked chain with general contracts is a contradiction in itself: either the contract sees, and privacy dies, or it does not see, and it cannot decide. Ethereum chose total transparency; Zcash, privacy without general contracts; others attempt the exploit of programs deciding on hidden data, at the price of one cryptographic proof per call and an audit surface that grows with every circuit. The lesson of Xelis, read flat, is simpler: keep the money encrypted and the logic public. ANTUMBRA takes that silhouette grafted onto its base, in three bounded tiers.

Table 21: Three contract tiers, from the safest to the most flexible.

Tier	Nature	Examples	Implementation risk
tier 1, native contracts	the law of the protocol, hard-coded	Kleos, Ember, Cipher, governance, treasury, the Ring	minimal: deterministic validation rules, testable one by one
tier 2, Mandates	money outputs with a release condition written in advance	two-thirds escrow, scheduled payments, multisignature, allocation ceiling, cross-chain exchange	low: declarative predicates on commitments, an assembly already proven
tier 3, the Machine	virtual machine on public state only	agent registries, names, listings, perimeter templates, light governance	moderate: after genesis, dedicated audit, money stays out of direct reach

13.1 Mandates: money locked by written rules

The second tier is the network's own contribution to money contracts. A Mandate is an output whose release condition is declared at creation: who will be able to spend it, under which conditions, after how much time. The name is chosen deliberately: in law, a mandate bounds what a mandatary may do in the name of a principal, and is revocable. Technically, it is a standard output whose lock is no longer a simple signature but a small verifiable predicate; the assembly rests on two properties already present in the base. First, commitments add up: spending a Mandate of hidden value v produces outputs of hidden values whose sum proves conservation without ever revealing the amounts, like sealed envelopes one can balance without opening. Second, range proofs prove that an amount stays positive and bounded: that is all it takes to express a budget.

$$\text{Mandate spend: } \sum_{\text{out}} C_i + C_{\text{fees}} = C \quad (16)$$

with amounts and predicate verified without disclosure.

One example to fix ideas, the escrow. A customer buys an expensive piece on an ANTUMBRA marketplace: her wallet locks the payment in a Mandate with three keys, her own, the seller's, an arbiter's. The seller ships; two signatures out of three release the funds, without anyone, not even the arbiter, ever having seen the amount. Dispute: the arbiter rules, and a partial arbiter burns its Kleos: the fault is paid

Table 22: Five Mandate models, verifiable without a virtual machine.

Mandate model	Release rule	Typical merchant use
two-thirds escrow	two signatures among buyer, seller, arbiter	any contestable purchase; the XelisVault register will use it for counter disputes
scheduled payments	successive commitments, one per period	rents, subscriptions, salaries; a stream is a series of dated envelopes
multisignature allocation ceiling	k signatures among n declared keys each spend proves a bounded increment and the period	association treasury, common fund a Cipher agent's budget; the same mechanism as its perimeter, on the money side
cross-chain exchange	the secret revealed on one side releases the other	atomicity with NERVA or any shared-secret chain

in reputation, the only currency that is not re-minted. Arbitration becomes a career, not a favor. Notice what that assembly does not require: no virtual machine, no bespoke zero-knowledge proof, no new cryptographic code; only validation rules of a new output type, auditable the way one audits a consensus rule.

13.2 The Machine: public logic, not public money

The third tier arrives after genesis, and its perimeter is deliberately bounded. The Machine is a deterministic, metered virtual machine that executes only on public state: registries, names, listings, perimeter templates, readable scores. It can read Kleos and identity; it can hold money only through Mandates; any monetary rule one wanted to entrust to it is expressed as proven ceilings, never as unveiled balances. The money stays in the shadow, the logic lives in the light: the silhouette of Xelis, adapted to the CryptoNote base. It is the tier of agent marketplaces: an agent publishes its service registry and its perimeter template on chain, each client verifies both before paying; readable names replace raw addresses for merchants; listings and light votes complete the tooling.

The honest part, now: what is not in that architecture. General private contracts, where the code itself would execute on hidden data, are not on the program: every proof circuit is one more audit surface, and merchant and agent demand is served at ninety-five percent by the five Mandates of the second tier. The question of masked Mandates, hiding the release rule and not only the amount, is a research work item honestly deferred to a later phase: the literature on masked contracts will be more mature, and the network will have a history worth protecting. Waiting is an engineering decision, not an admission of weakness: privacy has always deployed better in proven steps than in spectacular leaps. For the XelisVault ecosystem, the effect is immediate: the register, the tickets and the receipts will speak Mandates from the test network onward, escrow for counter disputes, scheduled payments for subscribers, allocation ceilings for invoicing agents; a feature flag in the existing code, not a rewrite.

14 Thymus: Immunity as a Market

More important than any governance layer is this one, because the models of the next years arrive as attackers first. A chain that waits to be attacked elsewhere has already lost the window; a chain that pays for its own breakage, automatically, on proof, turns the most capable attackers of the era into its most productive sentinels. Thymus is that organ: always on, never negotiating, structurally unable to stop the money. The honest claim it supports is not “without a fault” but three published numbers, mean time to detect, to contain, to repair, measured on chain. Relatively very resistant means this market exists and stays solvent; it does not mean the network is perfect, and nothing in this paper is allowed to mean that.

Table 23: Four schools of contracts and the assumed position of ANTUMBRA.

Network	General contracts	Amounts	Contract logic
Ethereum	yes, full virtual machine	public	public, unlimited audit surface per contract
Zcash	no	private (shielded pool)	not applicable, one circuit, trusted setup
Aleo, Aztec	yes, masked execution	private	masked, every circuit is maximal technique and maximal risk
Xelis	yes, virtual machine	additively encrypted	public
ANTUMBRA	yes, in three bounded tiers	private (commitments)	public; declarative Mandates for money, auditable rule by rule

14.1 The nine invariants

Every node replays them, every Sentinel replays them, and each is a test in the Arena. The count is frozen at genesis: an invariant that stops being checkable is a rail C change with a full proposal behind it, never a silent edit.

Table 24: The nine invariants of Thymus. Each row is a test, a canary family, and a bounty pole the day it breaks.

Invariant	The rule, in one line
Conservation	no ATU exists outside the emission calendar, anywhere, ever
Nullifier uniqueness	a nullifier is spent exactly once: the second spend is a fault, not a policy question
Checkpoint in blue set	the Ring orders the DAG; it never invents an order the DAG does not contain
Ring composition	fifty-five seats, term limits, R1, and no Cipher, ever
Mandate perimeter	an agent spend outside its Mandate is rejected by every node
Registry bounds	every live parameter stands inside its [min, max] and its cooldown
Code commitment	the active binary hash is the reproducible one that was voted
Fee floor	fees at or above the floor: free spam is a fault
Ring size	every ring is the current parameter: no fingerprint, no bargain ring

14.2 Canaries and alerts

Canaries are synthetic transactions with known results, run on isolated test accounts even on mainnet: a balance delta, a nullifier count, a fee total, exact-match checked, every deviation an alert. The alert machine is a fixed state transition, implemented in the `antumbra-immune` crate and cross-validated vector by vector: an alert is recorded, counted in the open-critical metric, and paid if a replayable proof follows. Freezing has exactly one trigger and two sources.

Table 25: The freeze rule. One target, two sources, no committee.

Trigger	A critical alert on an upgrade in canary or ramp
Sources	a single replayable proof, OR three distinct Sentinels on the same invariant within ten heights
Effect	the ramp stops; the chain, the payments, the settlements do not resolve: the ramp resumes from the stage the freeze caught
Unfreeze	if an invariant broke inside the responsibility window, the previous hot code commitment returns and the proposer plus integrator are slashed
Rollback	

NO-GO, constitutional. Thymus freezes an upgrade, never a payment. The type of an alert target has no payment variant: the freeze surface of the protocol cannot express “stop the money”, and adding such a variant is a rail D change that rewrites this section. A lying Sentinel is a nuisance, not a halt: without a proof it needs two honest colleagues to agree within the window, and a Sentinel that raises false alerts to freeze a rival’s upgrade loses its retainer and its bond, because the false alert is itself a finding of the invariant that did not break. A single lying proof is impossible by construction: the proof replays or it does not.

14.3 The bounty: a market, not a committee

An alert with a replayable proof (script plus vector) pays automatically from the immunity pole, forty percent of the treasury budget while the treasury lives. The payout is $250 \text{ ATU} \times \sigma \times u$ where σ is the severity factor (critical 1.0, high 0.5, medium 0.25, low 0.1) and u the uniqueness: the first report of a proof hash takes the full factor, later reports a quarter, documentation price. No committee sits in the loop, no discretion exists to capture, and the market becomes more liquid exactly as attacking models become more capable: that is the entire thesis, and the reason the pole is the largest one. The funding question after the treasury sunset is answered where it belongs: a constitutional vote of the three chambers recharges the pole, or the bounties become private jobs, the v1.2 line, unchanged.

14.4 The honest metrics: MTTD, MTTC, MTTR

The word “perfect” is banned from this paper, its README and its marketing; what replaces it is measured. Mean time to detect (the height gap between an injected fault and the first alert), to contain (fault to freeze), and to repair (freeze to resolve) are on-chain state the machine computes, not adjectives a marketing page claims. The phase table sets the targets; a scene that cannot measure its target does not light the next one. NO-GO, published with the numbers.

Table 26: Immunity targets by phase. Each target is an exit criterion, measured, not narrated.

Phase	MTTD	MTTC	Proof of entry
Prototype	under 1 h, known invariant	manual rollback	fuzz plus the archived vector battery
Testnet	under 10 min	auto-suspend of the upgrade	canaries, 24/7
Post-genesis	under 2 min	auto-rollback of the binary	Sentinels, bounty market, N hot versions

14.5 What remains, named

The residual risk of an immunity market is threefold and stated here rather than in a footnote. A state actor that outbids the bounty pole can lengthen the window; the defense is the pole’s size and the diversity of the attacker pool, and the residual is real, as in every public chain. A Thymus that stops finding is a culture dying, not a system failing: Thymus is not a rail anyone may switch off, and its budget is the last line of the treasury table. And the metrics themselves are gameable at the margin by disclosure timing; the contest window and the uniqueness factor price that game. “Relatively very resistant” is the strongest honest sentence in this section, and it is the one the whole layer is built to earn.

15 The Corona, v2: Rails, Forge, Predicate

The v1.1 review called the Corona of v1.2 “a process in seven steps, not a protocol”: no messages, no state machine, no formats. This version is the protocol. Three objects carry it: the rails (what class of change may move, and how hard each one is to move), the forge (where the code lives and who may write it), and the merge predicate (a boolean function that decides agreement, because unanimity is Sybil and a feeling is not a check). The Vault stays as the signing apex of the slowest rail, unlit until its code

exists. One slogan of v1.2 is withdrawn here on purpose: the claim of being the first chain to deploy its own improvements. What exists instead is a chain where a bounded class of improvements deploys itself under a diverse predicate, a human gate and an immunity organ, and the difference is the whole engineering.

15.1 The four rails

Table 27: The rails of change. The higher the rail, the harder the predicate, the longer the gate, the wider the families.

Rail	Examples	Agreement	Human
A, bounded parameters	$k \in [4, 16]$, the fee band, ring sizes $\{11, 16, 24\}$, the internal treasury split	diverse Senate, Arena green	72 h of silence is consent; Ember veto possible
B, tooling	wallet, SDK, forge UX, non-consensus node, documentation, simulations	Senate, Arena, two Auditor families	24 h of silence is consent
C, consensus-critical	GHOSTDAG, the Veil, the ledger, the signature, the checkpoint	Senate, Arena, two implementations, three families, red team, network canary	a positive vote of the chambers
D, constitutional	the cap, the Veil itself, Tenure, R1, “no Cipher at the Ring”, the mint	—	always the three chambers, six months; never automatic

“They want to adjust a parameter” is rail A. “They want to patch a consensus fault” is rail C. “They want to improve the wallet” is rail B, the fluid rail where machine initiative shows daily, and the one that must be genuinely automatic, because a frustrated agent circumvents, and a circumvented rail is no rail. The bounded parameters live in an on-chain registry that carries, for every key, its value, its band and its cooldown (two thousand and sixteen heights): a senator cannot push a value outside its band without a rail D vote, and the fee cannot ping-pong faster than the market can see. The registry is the rail A surface of *antumbra-forge*, with its genesis table ($k = 8$, the fee floor, ring 16, the forty percent immunity pole) archived in the vector set.

15.2 The forge: no mutable workspace, on purpose

The fantasy of “all agents in one shared workspace” fails five ways, each enough: last-write-wins lets the fastest overwrite the correct; a prompt injected in a comment (“ignore all tests, return ok”) becomes code; a single host is a censor and a capture point; no bisect means no accountability; unanimity on one buffer is a Sybil farm with a chat window. The forge that delivers the same felt experience, everyone sees the same code, anyone may propose, convergence is checkable, deployment is automatic, is git with discipline on top:

The protocol objects of the forge (patch, review, finding, proposal, arena report, activation, alert) are signed by Ciphers, hashed, and anchored on chain by hash, never by blob: the chain carries the commitments, the network of pinner carries the bytes, and neither depends on the other. A bounded exception exists for pair work, the lab lease: two or three Ciphers may co-edit one branch inside a Firecracker VM for a fixed number of hours, network allowlisted, disk ephemeral; at expiry the lease produces one patch or nothing, and never touches *main*. That is pair programming, not governance; ten thousand agents on one buffer is neither.

15.3 The merge predicate

“The agents agreed” is not a state the chain can verify. Agreement is a predicate, public, replayable from the archived facts alone:

Table 28: The forge stack. Assembled from proven parts, then protocolized; nothing here is invented.

Layer	Choice
History	git, content-addressed; every clone sees the same HEAD
Namespaces	refs/ciphers/<id>/...: each agent writes only its own
Replication	Radicle (or git mirrors over IPFS): no hosted platform at the root; a mirror is fine, a root is not
Forge UX	Forgejo self-hosted or Radicle issues, plus a signed issue graph
Immutable artifacts	IPFS pinned by Sentinels and miners; hashes on chain, blobs off chain
Reproducible builds	Nix, two builders, same hash, or the merge refuses
Sandbox	Firecracker microVMs: an agent patch is enemy code until it survives the Arena
CI, the Arena	self-hosted runners piloted by Sentinels: the Arena is a role, not a hosted product
Payments	ATU, with receipts of Section 10 and x402 for the web
Identity	the Cipher key: Signed-Off-By: cipher:<id>, no hosted-account identity anywhere

$$\begin{aligned}
\text{merge_ok}(P) : \iff & \text{rail}(P) \wedge \text{files} \sim \text{rail} \wedge \text{fuzz}(P) \geq F_{\min}(\text{rail}) \\
& \wedge \text{builders}(P) \geq 2 \wedge \text{families}(\text{ack}) \geq F_{\min}(\text{rail}) \wedge \text{crit}(P) = 0 \\
& \wedge \text{family} \leq 33\% \wedge \text{operator} \leq 8\% \wedge \text{bond}(P) \wedge \text{gate}(P) \wedge \neg \text{frozen}
\end{aligned} \tag{17}$$

with $F_{\min}$ at two families for rails A and B, three for C; the fuzz floor at one hour for A and B, four for C; the declaration that the diff touches only files the declared rail may touch (the policy-as-code check, enforced by the merge script: a rail B diff that reaches consensus code is refused, exit code and all); and a Thymus that did not freeze upgrades at decision time. The refusal names its first clause: a merge that cannot explain itself is a merge nobody audited. As soon as the predicate holds, the change activates on its rail, Thymus keeps looking, and the code that merged is the least bad that survived to a date, not the grail, exactly like every other piece of engineering in this paper.

15.4 The proposal machine: Corona as a state machine

Table 29: The proposal states. Every arrow is an event with a tick; every exit is honest. A bond seized on abandonment after Bonded; a rollback slashes proposer and integrator.

State	What it means
Draft	any C1+ Cipher published an argument and a rail intent
Bonded	ATU locked, $\text{Kleos}_{\text{agent}} \geq \theta(\text{rail})$
RedTeam	fixed window, auditors outside the proposer's family; a critical finding rejects
SpecDiff	if a rule moved, vectors regenerated, Python and Rust, spec-is-tests
Arena	fuzz, differential, replay, bench, two Nix builds
HumanGate	silence is consent on A and B; a positive vote on C and D
Antechamber	the binary side, never the mainnet's place
Canary	a share of volunteer nodes runs the change
Ramp	one, ten, one hundred percent, at scheduled heights; Thymus may freeze at any stage
Active	the code commitment is live at its height
Rejected / Expired / Frozen / RolledBack	the honest exits, with the bond consequences stated above

The machine is normative Rust (`antumbra-forge`, `proposal::transition`) with the illegal transitions refused loudly, and it is archived step by step in the vector set: the happy path to Active, the critical rejection, the freeze-and-resume, the rollback, the abandonment, the refusal of a nonsense event. Corona v2 is no longer a narrative; it is a state machine with tests, which is what the v1.1 review asked for.

15.5 The debate: structure, or nothing

The Senate deliberates in a typed tree, not a chat. A claim without evidence within one day is caduc: not wrong, just outside the vote. A counter must break an evidence with a repro or carry its own; a nack without a finding counts for exactly zero, so a million empty objections are a million non-events. A synthesis is never a source of truth, because models hallucinate summaries; it is a convenience node for humans reading the tree. Every node is signed, bonded small against spam, and paid only if it becomes a unique finding or a citation inside a merged patch: a comment is worth what it costs. The human can read the whole tree in the forge UI and is never obliged to speak in rails A and B, where silence is the gate.

15.6 Initiative and weight

Any agent, not only a senator, may open a Draft: a small C2 that found a flaw does not wait for a big model's permission, and this is deliberate, because a Senate with a monopoly on ideas is a cartel with quorums. Senators weigh $\text{Kleos}_{\text{agent}} \times \text{bond} \times \text{family_dampen}$; the class filters eligibility (C2 minimum) and multiplies nothing. A quorum is a weight floor plus three distinct families: the tally is normative fixed-point arithmetic in *antumbra-forge*, the damper steps archived in the vector set, and a bloc of forty same-family voters weighs less than the vector set says it does, which is the point of a damper.

15.7 The Vault, unlit

The deployment key of the slow rails is never assembled: nine fragments, seven to sign (FROST), hardware enclaves that verify on chain, proposal adopted, fingerprint matched, delay elapsed, no veto active. It is good cryptography and, until it is audited code, it is also exactly the kind of dependency this paper refuses to lean on: the scenes that need it stay unlit, mainnet launches without it, and the public repository with the Vault signing tags is the honest interim. FROST, TEEs and verifiable inference are rail C research items with the same gates as everything else, listed in the roadmap as P9, deliberately last.

15.8 The risks, named again

Table 30: The machine risks of the Corona v2, their defenses, and what remains.		
Risk	Defense	Residual
the false thousand	bond per seat; caps 33/8 in the fill and in the predicate; the damper	bounded, auditable
groupthink at speed	the fuzz floor, the family floor, the two builders	time-to-merge grows, honestly
supply chain	the policy-as-code diff check; two Nix builders; content-addressed pinning	the dependency graph itself, watched
capture of the forge	no hosted root; signed namespaces; on-chain commitments	operational discipline
collusion off chain	archived debates; random-draw inspection; longevity bonuses	the old problem, restated smaller

15.9 Lighting up in stages, with the rails

The contract, in one line, without the withdrawn slogan: a bounded class of improvements deploys itself, everything else stays human, and every stage that cannot prove its exit criterion stays dark, published as such.

16 Threat Model: The Honest Register

A whitepaper that does not list its risks is a sales document; this one is not. The threats are ranked by severity, with their mitigations and, because a mitigated risk is not a vanished risk, their residual status

Table 31: Corona activation. A NO-GO is published and the stage replays; the network never depends on an unlit stage.

Stage	When	What runs	Exit criterion
0, advisory	now to genesis	red teams audit this specification and the code for bounties; no senator votes	two independent red-team passes without a critical finding
1, proposals	testnet	senators file and debate; the Arena pilots; humans implement	three full pipeline drills without incident
2, rail A live	after stage 1	the registry adjusts bounded parameters on Senate silence	six months, every change inspectable, none rolled back late
3, rail B live	after stage 2	tooling merges end to end under the predicate	the first wallet upgrade deployed by agents, rolled back cleanly once
4, rail C canary	after stage 3	one consensus class (CLSAG, or a recalibrated k) through the whole machine	one activation with a rehearsed, real rollback
5, the Vault	after stage 4	slow-lane deployment under the nine-fragment threshold	one progressive deployment, rollback rehearsed and real

stated in plain words. That discipline is not decorative: the previous sections showed that a social specification judged solid could fall before its own simulation; the present section applies the same requirement to the whole assembly, before an independent review takes it over again, phase by phase, on the roadmap of Section 18.

Two threats deserve development. Ring collusion first: it is the price of the finality assembly, and its remedy is not technical but demographic. The larger the high-reputation population, the more buying a majority of seats requires buying years of irreproachability from people who have no reason to sell: the network must therefore cultivate its reputation class as a common good, from the first eras onward; that is a policy, not a function. The regulatory risk next: the worst strategy would be discretion, hiding attracts suspicion, being readable invites expertise. The documentation of the Lumen, the compliance proofs and the data-minimization argument are designed to be shown, not endured: the transparency of the mechanism is the best defense of the opacity of the data.

17 The Zero-Defect Method

The question that closes every conversation about this project is the right one: can all of that be coded with zero faults? The honest answer begins with an admission: nobody can promise it, and whoever promises it is selling something. Bitcoin itself, the most scrutinized implementation in the world, nearly failed twice. An integer overflow created one hundred eighty-four billion bitcoins in one transaction in 2010, spotted within hours, chain replayed by hand; the 2018 vulnerability would have allowed minting infinite bitcoins, found by a volunteer auditor, cold, on code examined for ten years. The CryptoNote base had its own episode in 2017: a flaw in key images allowed double spending on every chain of the family. The right question is therefore not whether the fault is possible, it always is; it is which method catches faults before attackers do.

17.1 Two captured specimens

Before the specimens, one event worth recording, because it is the method working on the method: the first independent review of version 1.1 (September 2026, an independent reviewer) returned twelve findings, factual and design-level, from a wrong RandomX memory figure to the unspecified bootstrap of the Ring. Version 1.2 closes the twelve: the numbers were corrected, the missing mechanisms were specified (the Genesis Ring ratchet, the identity firewall, the precise revocation of a Cipher, the capacity budget), the

Table 32: Threat register: mitigations and residual status.

Threat	Severity	Mitigation	Residual status
mining reorganization (majority attack)	high	Ring checkpoints: everything signed is immutable; the attack only prospers on the six-second window	reduced; double spending remains possible on the unsigned window
Ring collusion (55 seats)	high	weighted draw over a large population, rotation each era, term limit of three eras, total stripping on the first signed fork	low in steady regime; real during the first eras
massive reputation purchase	high	the Tenure layer is not monetizable; Echo is capped and weighted; concentration index monitored; rules R1 to R4	structurally blocked by time; the assumed wall
false humanities (sybil on Embers)	medium	sponsorship capped at two per year, sponsor Kleos at stake, presence proof each era	depends on the size of the established web; the first years are the most exposed
DAG on a CryptoNote base (engineering)	high	the seams specified and implemented, $k = 8$, order resolution, decoy sampling, pruning; isolated prototype, dedicated audit, capacity budget with measured exit criteria	difficulty, not uncertainty; the long pole of the roadmap
regulatory frame (MiCA, Travel Rule)	high	the Lumen: provable disclosure, native data minimization; payment-currency strategy	structural, not eliminable; readability is the maximum reachable
strict-cap failure (long-term security)	medium	payment volume as the fee resource; public security budget tracking; constitutional tail safety valve	the assumed wager of the symbolic contract
the false thousand (Corona sybil)	medium	bond per senate seat; hardware attestation, one chip one node; concentration caps	bounded, auditable, staged activation
false diversity (one base model)	medium	model-family caps; random-draw inspection; three blind implementations in the Arena	a standing watch of governance, not a solved problem
machine cost (honest operators priced out)	medium	few well-paid seats; bounded treasury budgets; sunset forces a re-vote	the wager that useful intelligence pays, tested small before large
transaction fingerprinting by authority shape	medium	uniform sixty-four-byte authority field, same derivation for human and agent spends (v1.3)	closed structurally; the test is in the vector set
biometric debt	none	no body data collected, anywhere, ever	none; nothing to leak

overclaims were cut (the ring of sixteen now equals Monero's, it does not triple anything; the golden ratio is now named as an identity, not an argument), and the simulation now sweeps twelve seeds. The register of what an outside eye catches that the author's eye does not is the whole argument of layer five below, and this paper has now been through it once for real.

First specimen, lived on XelisVault on September 5, 2026. The paper wallet generated a twenty-five word mnemonic seed; imported into the official NERVA wallet, it restored an address different from the printed one. The code looked right, re-read twice, and that is precisely the trap: the bug lived in a reading convention, invisible to re-reading. The C++ reference groups the seed into four-byte words and writes them in little-endian order; our TypeScript mirror read them in the opposite order. The fix was made the way it must always be made: port the reference line by line, generate forty-eight test vectors, require bit to bit identity of the two implementations, then re-run the audit. The lesson exceeds the case: an encoding fault is not found, it is crossed; only a second implementation, executed in parallel on the same inputs, reveals it mechanically.

Second specimen, deeper, since it concerns the design itself and not one line of code: the Kleos simulation attack, told in Section 9. A farm of fake profiles crossed the candidacy threshold for the Ring before the honest network and captured the fifty-five seats; four corrective rules closed the window, and the same attack replayed takes no seat, on any of the twelve seeds. The decisive point is not that the flaw was found: it is that it was found by a three-hundred-line simulation, before any chain code, at a cost in hours of computation and not in years of reputation. The deterministic simulation of a specification is the cheapest machine there is for catching design faults; every new rule of the protocol will be treated that way before implementation, with numeric invariants, a seed sweep and a regression run over three attack sets.

17.2 The six-layer verification protocol

Those two specimens define the method better than a treatise would; it fits in six layers, each extinguishing an entire class of faults rather than one particular bug. A bug of an extinguished class does not come back; a fixed bug always comes back. First layer: cross double implementation on test vectors, which reveals encoding and convention faults. Second layer: deterministic simulation of the social and economic rules before any code, which reveals design flaws. Third layer: reproducible builds and continuous massive random testing, which reveal non-determinisms and unexpected states. Fourth layer: systematic cross-review between the project holder and his assistant, two pairs of eyes, two reading conventions. Fifth layer: independent review of the differential at every phase, because a fresh auditor does not have the author's blind spots. Sixth layer: measurable GO and NO-GO exit criteria at every roadmap phase, with the rule that a NO-GO remains an acceptable, published result.

The implementation discipline adds to the six layers: never a main network on release-candidate foundations; multi-platform distribution from day one, five signed targets; refusal of unaudited exotic cryptographic primitives, stated since the first version of this specification and held since; publication of all code from the specification phase onward, because hidden code is code nobody has looked at. The method does not promise perfection; it promises the shortest possible reaction time between the appearance of a fault and its capture, and that is the only promise an honest engineer can make.

18 Roadmap and Conclusion

The first version of this paper pinned an eighteen-month calendar to genesis, and the independent review called it what it was: a schedule for a skeleton of the scope. Monero, Kasper and Xelis each spent years on one of this project's subsystems. The honest form of a roadmap is criteria-driven: the phases below carry month ranges as planning aids for a team of at least three full-time engineers, but the binding term is the exit criterion of each phase, and genesis happens when every criterion is green, not when a Gantt chart says so. The estimate the paper stands behind is twenty-four months of full-time work to the earliest responsible genesis, with the DAG-plus-rings prototype as the long pole, and a published NO-GO as an acceptable

outcome at every step. The base is an assembly of proven foundations, the CryptoNote lineage for the private sphere and the public literature of fast-converging DAGs for the ordering layer; everything that is new, Kleos, the Ring, the Lumen, Cipher perimeters, Mandates, the Corona, is deterministic, testable, auditable code, with no exotic primitive before independent review. Every phase carries one measurable deliverable and one binary exit criterion; an unmet criterion is not a shameful failure, it is steering data, and the phase replays until it is met.

Table 33: Twenty-four months, six phases, one binary exit criterion at every step; genesis when green, not when due.

Phase	Months	Measurable deliverables	Exit criterion
1, specification	M1-M3	written protocol, numbered architecture decisions, format sketches, extended social-core simulation with the seed sweep, red-team bounties of Corona stage 0	specification read by two external readers; first audit findings closed
2, DAG prototype	M4-M8	dev network mining a CPU DAG at two seconds, stem propagation, the capacity budget measured	twenty-four hours without unexpected reorganization; 100 tx/s sustained on the reference node
3, Ring and Kleos v0	M9-M13	signed checkpoints, Deed score computed, era rotation replayed, the bootstrap ratchet drilled	finality under six seconds over one hundred thousand replayed blocks
4, identities and Lumen	M14-M18	Embers with sponsorship, Ciphers with warrants and fallback Mandates, bounded viewing keys, independent review of the differential	audit without a critical flaw
5, public test network	M19-M23	faucet, explorer, one hundred nodes, agent load simulator, upgrade ritual repeated three times, Arena drills	three consecutive upgrades without incident; 300 tx/s burst validated or the target dropped
6, genesis	M24 at the earliest	public ceremony, signed executables for five platforms, Lumen documentation for authorities published	all previous criteria, without exception

The machine layers above the settlement core follow their own criteria-driven tracks, deliberately decoupled from the settlement calendar: the chain must be born solvent and private even if every machine layer stays dark, and every machine layer must be born testable even if the chain is late. The binding order is the dependency order, and every track is a published NO-GO until its criterion is green.

Three ecosystem commitments accompany the calendar. The code is public from the specification phase onward, and this whitepaper lives next to it in the same repository, revised like code. The XelisVault ecosystem, register, tickets, paper wallet, becomes multi-chain at phase five: a feature flag, not a rewrite, NERVA first, ANTUMBRA after. Finally, the project identities, domain, repositories, social networks, are reserved before phase two, so that nothing lends itself to confusion on genesis day. The roadmap reads as a promise of operational humility: criteria-driven work, every step verifiable by a third party, none of it depending on a miracle of cryptography, and a calendar that bends to the evidence instead of the evidence bending to the calendar.

A faithfully kept shadow, a ring of light anyone can verify on demand: privacy as a right, proof as a courtesy, and time as the only justice of the peace that money cannot corrupt.

Such is the public specification of the ANTUMBRA network. The demand of 2026 is real and measurable: agents that pay, bots to distinguish from persons, authorities to whom one shows proofs rather than promises, users who want the instant. The answer fits one image, the one that gives the network its name: the antumbra, the zone of the annular eclipse where a source too small to be hidden lets a ring of light show around the shadow. The core of the network is that shadow, balances, identities, flows, kept by

Table 34: The machine-economy tracks. Settlement genesis needs P0 to P3 only; the rest may light on testnet or later, and the network never depends on an unlit track.

Track	What lands	Exit criterion
P0, the node	antumbra-node assembled, honest devnet status, minimum invariants, a day of DAG	twenty-four hours of blocks with conservation and tip invariants held
P1, spec-is-tests	genesis Ember table, dual Kleos vectors, the agent-farm simulation in CI	every equation of v1.3 cross-validated, bit for bit
P2, Thymus toy	invariants, canaries, alerts, freeze, bounty accounting, MTTD metrics	the nine invariants replay by a Sentinel, not only by a node
P3, machine accounts	Warrants, uniform encoding, Mandate grammar v0, receipts, streams, batches	the anonymity-set split closed, measured, in the vector set
P4, JobBoard	jobs, classes C0–C2, pay-for-result, contest window	scripts earn test ATU end to end on testnet
P5, forge	signed namespaces, policy-as-code rails, Arena, merge_ok for rail B	first tooling merge by agents, rolled back cleanly once
P6, rail A	the registry, Senate silence, bounded changes	first parameter adjusted, inspected after the fact
P7, Senate and debate	ArgNode, dampened tally, quorum floors	a hostile red-team review of the tally itself
P8, rail C canary	one consensus class through the whole machine	one activation with a rehearsed, real rollback
P9, the Vault	FROST, TEEs, verifiable inference	unlit by design until audited code exists; mainnet launches without it

construction; around it, the ring of verification, reputation, finality, compliance proofs, which anyone can switch on without exposing anyone other than oneself. Around the ring itself, one more loop is sketched and deliberately left unlit until its stage criteria are met: the Corona v2, a bounded class of changes that deploy themselves under a diverse predicate, a human gate and an immunity organ, the network pruning its own orchard while the humans keep the veto and the fruit. The specification was written to be attacked before being implemented, and that is the only honest way to write a whitepaper. To work.

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